

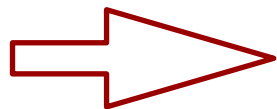
Peter Willendrup

Establishing the learning goals, a look at the programme

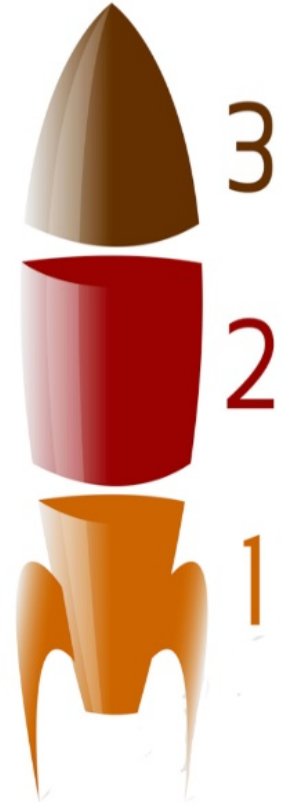
Learning goals:



1. Learn McStas basics
2. Build and operate simple instrument models, source + optics + sample + detector
3. Learn basics of instrument-optimisation and resolution calculations
5. Add Mantid / NeXus capabilities
6. Get a better idea of what you want to do with McStas, how to do it, how to get help
7. Get up-to-speed with latest developments and advanced features of McStas and related tools like NCrystal and McStasScript.



Enable your independent work with McStas





EUROPEAN
SPALLATION
SOURCE

School programme



2025 ISIS
McStas
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November 4th Beginners McStas	Time (GMT)	November 5th Instrument design	Time (GMT)	November 6th Advanced A	November 6th Advanced B
Welcome + setting learning goals 30 min McStas intro + general concepts 30 min McStas live demo / "Build-along" Responsible: Peter	9:00-10:30	30 min Polarisation 60 min Lesser known tips and tricks, including: - Tools for optimisation Responsibles: Peter	9:00-10:30	Advanced topics: NCrystal talk Using NCrystal in McStas & McStasScript Responsible: Thomas	Work on your own instrument w / help and guidance
Coffee Break	10:30-11:00	Coffee Break	10:30-11:00		Break
60min Guides and gravity: 20 min presentation 40 min practical Responsible: Mads	11:00-12:30	Time of flight resolution, talk + exercise Responsible: Mads	11:00-12:30	Advanced topics interactive: Write your first component Responsible: Peter	Work on your own instrument w / help and guidance
Lunch break	12:30-13:30	Lunch break	12:30-13:30	Lunch break	Lunch break
Choppers / rotating optics: 20 min presentation 25 min practical 45 min Sample + Union overview talk Responsibles: Mads + Peter	13:30-15:00	McStas -> Mantid, NeXus: 20 min presentation 20 min demo 50 min exercise Responsible: Peter	13:30-15:00	Advanced talk/demo: McStas Union for detectors Responsible: Mads	Work on your own instrument w / help and guidance
Coffee Break	15:00-15:30	Coffee Break	15:00-15:30	Coffee Break	Break
45 min Sample exercise 45 min Union exercise Responsibles: Peter + Mads	15:30-17:00	90 min McStasScript intro/exercise Session lead: Mads	15:30-17:00	Porting old comps and instruments to McStas 3 (and allow GPU execution) Responsible: Peter	Work on your own instrument w / help and guidance

Links to various infrastructure and services at:
<https://isis2025.mcstas.org>

School programme - day 1



November 4th Beginners McStas

Welcome + setting learning goals
30 min McStas intro + general concepts
30 min McStas live demo / "Build-along"

Responsible: Peter

Coffee Break

60min Guides and gravity:
20 min presentation
40 min practical

Responsible: Mads

Lunch break

Choppers / rotating optics:
20 min presentation
25 min practical
45 min Sample + Union overview talk
Responsibles: Mads + Peter

Coffee Break

45 min Sample exercise
45 min Union exercise

Responsibles: Peter + Mads

Intro lecture, general principles,
sources + monitors,

Lectures + "recipe" exercises

Sample-lecture, including "advanced McStas
grammar", sample exercises

In "cookbook" sections,
think ahead toward
your own project:

- * Which neutron source
- * What optics
- * What sample

- K.I.S.S. for now

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School programme - day 2



November 5th Instrument design

30 min Polarisation
60 min Lesser known tips and tricks, including:
- Tools for optimisation

Lectures on polarisation
and instrument optimisation
technicals

Responsibles: Peter

Coffee Break

Time of flight resolution, talk + exercise

ToF resolution, talk + exercise

Responsible: Mads

Lunch break

McStas -> Mantid, NeXus:

20 min presentation

20 min demo

50 min exercise

Responsible: Peter

Mantid-howto,
lecture and demo

Coffee Break

90 min McStasScript intro/exercise

McStasScript intro/exercise

Session lead: Mads

School programme - day 3, advanced stuff + work on own project



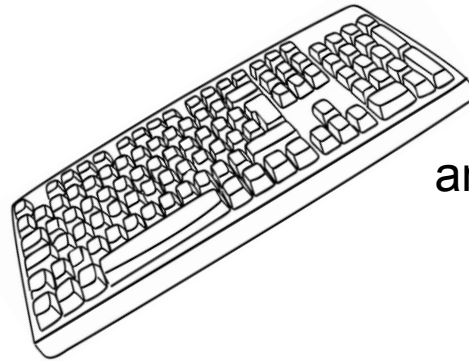
November 6th Advanced A	November 6th Advanced B
Advanced topics: NCrystal talk Using NCrystal in McStas & McStasscript Responsible: Thomas	Work on your own instrument w / help and guidance
	Break
Advanced topics interactive: Write your first component Responsible: Peter	Work on your own instrument w / help and guidance
Lunch break	Lunch break
Advanced talk/demo: McStas Union for detectors Responsible: Mads	Work on your own instrument w / help and guidance
Coffee Break	Break
Porting old comps and instruments to McStas 3 (and allow GPU execution) Responsible: Peter	Work on your own instrument w / help and guidance



For the exercise-based work-sessions

- You will benefit from working in pairs, $2 > 1$

- Take turns being the “coder”



and the “parallel processor”



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Let's get to it!

